

Experiment 4

Finding Multiplicative Inverse using Extended Euclidean Theorem

Code :

// C++ program for finding Multiplicative inverse using extended Euclidean Algorithm

```
#include <bits/stdc++.h>
```

```
using namespace std;
```

// Function for finding multiplicative inverse

```
int gcdExtended(int a, int b, int &x, int &y) {
```

```
    // Base Case
```

```
    if (a == 0) {
```

```
        x = 0;
```

```
        y = 1;
```

```
        return x;
```

```
    }
```

```
    int x1, y1;
```

```
    int gcd = gcdExtended(b%a, a, x1, y1);
```

```
    x = y1 - (b/a) * x1;
```

```
    y = x1;
```

```
    return x;
```

```
}
```

```
int findmuv(int a, int b) {
```

```
    int x = 1, y = 1;
```

```
    return gcdExtended(a, b, x, y);
```

```

}
int find_gcd(int a,int b){
    if (a == 0)
        return b;
    return find_gcd(b % a, a);
}
int main() {
    cout<<"Enter number of test cases: "<<endl;
    int t ;
    cin>>t;
    while(t--){
        int num, mod;
        cout<<"Enter a number"<<endl;
        cin>>num;
        cout<<"Enter a Mod val"<<endl;
        cin>>mod;
        cout<<"Finding the multiplicative inverse of "<<num
<<" in mod "<<mod<<endl;
        int g = find_gcd(num,mod);
        if(g == 1){
            int mul_inv = findmuv(num, mod);
            if(mul_inv < 0){
                mul_inv+= mod;
            }
            cout << mul_inv << endl;
        }
        else{
            cout<<"Multiplicative Inverse doesn't exist "<<endl;
        }
        cout<<endl;
    }
    return 0;
}

```

OUTPUT :

```
C:\Users\pc\Documents\Experiment 4.exe
Enter number of test cases:
4
Enter a number
3
Enter a Mod val
11
Finding the multiplicative inverse of 3 in mod 11
4

Enter a number
11
Enter a Mod val
26
Finding the multiplicative inverse of 11 in mod 26
19

Enter a number
23
Enter a Mod val
100
Finding the multiplicative inverse of 23 in mod 100
87

Enter a number
12
Enter a Mod val
26
Finding the multiplicative inverse of 12 in mod 26
Multiplicative Inverse doesn't exist

-----
Process exited after 124 seconds with return value 0
Press any key to continue . . .
```

Explanation :

- Added Three test cases one in which multiplicative inverse doesn't exist
- Second one in which Multiplicative inverse exists and is positive
- Third one in which Multiplicative inverse is negative first and then after taking mod comes out to be positive